

M0 — Feasibility Memo

Group: Demonstration (instructor's worked example) **Members:** AB, CD, EF **Date:** 22 September 2026

The paper we are locking

Kueh MTW, Chew NWS, et al. Body weight categories and fat distribution in relation to all-cause mortality among adults with metabolic dysfunction-associated steatotic liver disease. *BMJ Open* 2026;16:e113719. doi:10.1136/bmjopen-2025-113719

Open access (CC BY-NC). Built entirely on public NHANES data with NCHS linked mortality.

Our estimands, fixed for the term:

Scientific (locked) estimand — IV vs III. Among adults meeting FLI-based MASLD criteria, do **non-obese adults with high central adiposity** have higher all-cause mortality than **non-obese adults with low central adiposity** — and how robust is that to cohort construction, confounding, effect heterogeneity, survey design, and incomplete observation?

Replication estimand — IV vs I. The paper references its contrasts to Group I (obese, low central adiposity). We reproduce that, because reproducing it is what *Replicate* means.

Both are contrasts within one fitted model; re-levelling is reparameterization, not reanalysis. Every estimate we report says which estimand it belongs to. The locked contrast is the one the term interrogates, because it holds body-mass category fixed and so isolates central adiposity.

The six criteria

Each claim below is confirmed with a **specific pointer**: a file name, a variable name, or a number we counted ourselves.

1. Reconstructable data — met

Six NHANES cycles, all public, no data-use agreement required:

Cycle	Suffix	Cycle	Suffix
2007–2008	_E	2013–2014	_H
2009–2010	_F	2015–2016	_I
2011–2012	_G	2017–2018	_J

Download pattern: https://wwwn.cdc.gov/Nchs/Data/Nhanes/Public/<year1>/DataFiles/<COMPONENT>_<suffix>.XPT

Components we need: DEMO, BMX, BPX, TRIGLY, BIOPRO, GHB, GLU, HDL, TCHOL, ALQ, SMQ, BPQ, DIQ, MCQ, KIQ_U, PAQ, ALB_CR, RXQ_RX.

Mortality linkage: NHANES_<cycle>_MORT_2019_PUBLIC.dat from the NCHS linked mortality files (fixed-width, public-use).

Merge key: SEQN, present in every file including the mortality .dat.

We downloaded all six cycles and merged them before writing this memo — the funnel in criterion 4 is counted from our own merged file, not read off the paper.

2. A clear exposure and outcome — met

Exposure — a 2×2 phenotype, not a continuous variable:

- Obesity axis: $\text{BMXBMI} \geq 30$ (≥ 25 if $\text{RIDRETH3} = \text{Asian}$)
- Central-adiposity axis: $\text{WHtR} = \text{BMXWAIST} / \text{BMXHT} \geq 0.6$

giving Groups I (obese, low central), II (obese, high central), III (non-obese, low), IV (non-obese, high). Group IV is the paper’s headline.

Outcome — all-cause mortality: $\text{MORTSTAT} = 1$, follow-up time PERMTH_EXM , censored 31 December 2019.

Eligibility — MASLD by Fatty Liver Index ≥ 60 plus ≥ 1 cardiometabolic criterion. FLI is computed from LBXTR (triglycerides), LBXSGTSI (GGT), BMXBMI and BMXWAIST .

3. Plausible confounders — met

Measured and available on every cycle:

Confounder	Variable(s)
Age	RIDAGEYR
Sex	RIAGENDR
Race / ethnicity	RIDRETH3
Income-to-poverty ratio	INDFMPIR
Smoking	SMQ020 , SMQ040
Type 2 diabetes	$\text{DIQ010} + \text{LBXGH}$ (HbA1c)
Hypertension	$\text{BPQ020} + \text{measured BPXSY1-BPXSY4}$
Kidney function	LBXSCR (CKD-EPI 2021), ALB_CR (ACR)
Medication classes	RXQ_RX

A judgment we are flagging now, not at M1: several of these — T2DM, hypertension, CKD, prior MI — **could be descendants or intermediates under plausible causal structures, but their causal roles and temporal ordering are not established here.** The paper adjusts for all of them. We will reproduce that adjustment set because replication requires it, and we expect to argue at M1 that the crude-to-adjusted attenuation **cannot be attributed** to confounding control without a prespecified causal model supporting the adjustment set. Temporal ordering supports adjusting for age, sex, race, income and smoking without reservation.

4. Adequate size and events — met, with one caveat

Counted from our merged file. This is the chain to the paper’s own analytic file (the **full analytic file**), which requires a computable FLI but not a positive fasting weight, so the fasting-subsample step (DESIGN) and missing FLI inputs (MISSINGNESS) are fused into one row here:

Step	N
All NHANES 2007–2018 records	59,842
Age ≥ 18	36,580
FLI computable (TG, GGT, BMI, waist present)	14,989
Steatosis, FLI ≥ 60	6,376
MASLD (≥ 1 cardiometabolic criterion)	6,374
Mortality-eligible and linked	6,371

Criterion 6 keeps the two steps apart by counting inside the fasting subsample. Counted that way, the chain runs 14,962 adults sampled, 14,170 with a computable FLI, and ends at 6,048 mortality-eligible and linked records with a valid fasting weight. The two chains are different row sets, not a discrepancy.

On the 6,371 rows, group sizes are **391 / 4,585 / 787 / 608; 586 deaths**.

That is ample for the main analysis. **The caveat is Group I**, the reference group, at $n = 391$. A sex-stratified interaction analysis (P3) splits it further, so stratum-specific estimates in Group I will be imprecise and we should expect wide confidence intervals rather than treat them as null findings. We are proceeding, with that stated in advance.

5. Design variables — met

All present in DEMO and TRIGLY:

- WTMEC2YR — MEC examination weight
- WTSAF2YR — **fasting subsample weight**, which is the correct one here, because FLI requires fasting triglycerides
- SDMVSTRA — stratum
- SDMVPSU — primary sampling unit
- SDDSRVYR — cycle indicator

Because we pool six two-year cycles, the weight must be divided by 6. The published analysis appears to be **unweighted**, which is one of the improvements we expect to propose.

6. A genuine incomplete-observation problem — met

We claim only that the paper **can support** an incomplete-observation analysis. We do not claim the problem is large, because we have not yet measured whether it matters.

First, what does **not** count. Four kinds of loss shrink this cohort, and none of them satisfies criterion 6 on its own, because P5 does not address them with the methods this course teaches. We must be able to tell each apart from missingness, and at P5 we report it and scope the conclusion around it.

- **DESIGN.** Fasting laboratory measures come from a **designed** NHANES subsample, and WTSAF2YR exists precisely to account for that selection stage and its nonresponse. Their absence outside the subsample is a design feature, not by itself a missing-data problem or evidence of MNAR — and the syllabus is explicit that designed subsampling alone does not satisfy this criterion. Inference does remain conditional on the weighting assumptions.
- **ELIGIBILITY.** The restriction to adults removes records that were never in scope. That states who the paper is about; it is not attrition.
- **TARGET POPULATION.** The steatosis and MASLD criteria remove people without the disease. They are not missing — they are **not who the paper is about**.
- **OUTCOME ASCERTAINMENT.** Records that could not be linked or followed for mortality were never in a position to carry an outcome value: we count them, report them separately, and do not impute them. A record that *was* followed and still lacks an outcome value would be missingness, and belongs in the list below.

What does count, measured inside the fasting frame:

- **Item nonresponse on the exposure inputs.** FLI needs triglycerides, GGT, BMI and waist. Among the 14,962 adults in the fasting frame, **792 (5.3%)** lack a computable FLI. These are people who *were* selected into the subsample and still cannot be classified.
- **Residual covariate missingness** within the analytic cohort, on variables the paper's own Model 1 uses.
- **A checkable missingness mechanism.** Are the 792 who lack a computable FLI systematically different from those who have one? NHANES retains demographic and examination data on them, so the MAR argument P5 will need is checkable rather than merely assertable. This is a diagnostic on the missingness itself. The selection stage upstream of it is DESIGN — handled by WTSAF2YR and by a sentence naming who our estimate is about, not by anything we would impute.

We are not arguing from magnitude. 5.3% is not self-evidently serious, and it may turn out to change nothing. The criterion asks whether there is a real incomplete- observation issue that requires explicit

assumptions and can support a substantive sensitivity analysis. There is, and whether it matters is P5's job to measure rather than ours to presume.

Feasibility verdict

All six criteria are met. We are locking this paper.

The thing we already expect to be interesting is that the FLI eligibility index shares its inputs (BMI and waist) with the exposure definition — cohort selection and exposure definition are built from the same measurements. That is a candidate threat we will develop across P1–P5. Which threat actually dominates is a judgment for M1, once all five increments have been run; naming it now would be guessing.

Cover note

Contribution. AB: data download, merge, funnel counts. CD: variable mapping for criteria 2–3, confounder table, temporal-ordering argument. EF: design variables, missingness quantification, memo assembly.

External feedback. None received.

AI Use Statement. Tool and version: Claude (Opus 5), used in RStudio and in a browser. Used for: drafting the download loop across the six cycle suffixes, and for checking the NHANES variable names for GGT and creatinine against the codebooks. Not used for: the six eligibility judgments. Each was verified by us against the actual codebooks and the actual downloaded files, and every N in criterion 4 was counted from our own merged data. An AI's guess about what NHANES contains is not a feasibility check.